Ions and Bonding

The Nobel gasses(group_____) are a group of elements that have their outer most electrons shell full. This makes them very **stable.**

Other elements loose, gain or share electrons in order to have their outer most electrons shells full. They try to get their outer shells like a noble gas.
When the atom of an element loses or gains electrons it becomes a charged particle called an _____. _____ can have a negative or positive charge.
Metals tend to lose/gain electrons to get their outermost shell full.
Metals from _____ ions.
Non-metals tend to gain electrons and become _____ ions.

Practical

Background
Iron is a metal/non-metal. Metals tend to loose electrons to form positive/negative ions.
Sulphur is a metal/non-metal. Non-metals tend to loose electrons to form positive/negative ions.

Look at the sulphur and record your observations in the table below.
Look at the irons filings and record your observations in the table below.
Place about cm of sulphur in the bottom of a test tube.
Add about 1 cm of iron filings to the sulphur.
Shake the mixture up.
Put on safety gasses.
Do not put your head directly over the test tube.
Do NOT SNIFF the yellow gas produced. You will smell it soon enough.
Heat the mixture.
Record your results in the table.

Material	Colour before heating	Smell before heating	Colour after	Smell after heating
Sulfur				
Iron				
Irons sulfide				

Review of what we saw.
When the mixture was heated the iron atoms _____ electrons and the sulphur atoms _____ electrons. The irons became _____ ions and the sulphur became _____ ions.
The _____ sulphur ions where attracted to the _____ iron ions. Opposite charge attract.
Ions stuck together to form the compound iron sulphide.
This bonding where ions bond together due to a difference in charge is called ionic bonding.